MIL-H-43079F
14 October 1988
SUPERSEDING
MIL-H-43079E
15 June 1984

MILITARY SPECIFICATION

HOOD, ROCKET FUEL HANDLER'S:

IMPERMEABLE, LIMITED PROTECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a hood to be worn to provide protection against fuels and oxidizers.

1.2 Classification. The hoods shall be of one type, two classes, and one size (see 6.2):

Class 1 - Army
Class 2 - Air Force and Navy

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
L-P-535	-	Plastic Sheet (Sheeting): Plastic Strip: Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid
O-C-105	-	Calcium Chloride, Dihydrate and Calcium Chloride, Anhydrous: Technical
FF-N-836	-	Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat
FF-S-92	-	Screw, Machine: Slotted, Cross-Recessed or Hexagon Head
HH-C-576	-	Cork Sheet, (Gaskets, Sheets, and Strips)
UU-P-328	-	Paper, Manifold
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-F-495	-	Finish, Chemical, Black, For Copper Alloys
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-7254	-	Propellant, Nitric Acid
MIL-F-10884	-	Fasteners, Snap
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43062	-	Cloth, Coated, Nylon, Resin Modified Butyl-Coated, Acid and Fuel Resistant
MIL-P-51493	-	Paper, Chemical Agent Detector: M9

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-406	-	Plastics: Methods of Testing
FED-STD-601	-	Rubber: Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

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(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-1-3	-	Jig, Acid Testing No. 2
8-2-16	-	Hood, Rocket Fuel Handler's, Impermeable, Limited Protection, Class 1 US Army
8-2-24	-	Hood, Rocket Fuel Handler's, Impermeable, Limited Protection, Class II US Navy and US Air Force

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the document may appear in the sample in which case the document shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials and components (see 6.6).

3.3.1 Frame assembly.

3.3.1.1 Frame. The frame shall conform to Fibre Metal Products Co., part number 622, or equal, except the facepiece, holes for mounting the facepiece, and mounting screws shall be omitted (see 6.5 and 6.9).

3.3.1.2 Headband, cushion. The front half of the headband shall be provided with a leather cushion with a soft lining approximately 1/16 inch thick. The cushion shall attach to the headband by folding lengthwise around the headband.

3.3.1.3 Facepiece. The transparent facepiece shall be a polyvinyl chloride sheet conforming to composition A, type I, class 1 of L-P-535, optically clear and colorless with a high polish finish on both sides. A slight green tint shall be permissible provided that the tinting does not affect the optical clarity. The thickness shall be 0.040 ± 0.004 inch. The facepieces shall conform to the patterns specified in 3.5.1 and shall be permanently shaped to fit the frame.

3.3.1.4 Cork. The cork for facepiece reinforcement shall be cork - composition sheet, 1/8-inch thick, conforming to type I or II, A or B, class 1 or 2 of HH-C-576.

3.3.1.5 Screw and nut combination. The screw and nut combination for attaching the facepiece to the hood shall be as specified in 3.3.1.5.1 and 3.3.1.5.2 and shall be made of like material.

3.3.1.5.1 Screw. The screw shall conform to type I, style 9s, brass or aluminum alloy, size 6-32 UNC-2A of FF-S-92 and of sufficient length to provide a tight fit between the facepiece and the hood.

3.3.1.5.2 Nut. The nut shall conform to type III, style 15, size 6-32 UNC-2 , brass or aluminum alloy of FF-N-836.

3.3.2 Basic fabric. The basic material shall be resin-modified butyl coated nylon, acid and fuel resistant, conforming to MIL-C-43062.

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3.3.3 Strapping and front reinforcement. The gum strapping used to seal the seams and the front reinforcement shall be made from unsupported film of the uncured coating compound used to coat the fabric. The gum strapping and reinforcement shall be 0.012 to 0.025-inch thick and thoroughly clean before application. The gum strapping shall be 3/4 ± 1/8 inch wide and the front reinforcement shall be cut to pattern dimensions (see 4.4.1.1).

3.3.4 Cement. The cement used to seal the seams and the strapping over the seams shall be a curing-modified butyl base cement and shall meet the performance requirements specified in 3.3.9. The color of the cement, when dry, shall be approximately that of the coated fabric. As an alternative, a clear cement shall be used.

3.3.5 Loops and straps. The loops and straps shall be made from the basic coated fabric specified in 3.3.2. The straps and loops shall be coated on the lightly coated side with cement specified in 3.3.4, folded evenly lengthwise, and rolled to ensure good adhesion.

3.3.6 Labels. Each hood shall have a combination identification and instruction label conforming to type IV, class 15 of DDD-L-20 except that the Date of Manufacture shall appear on the last line of inscription on the label. The marking medium shall be white, yellow, or light gray. The label may be marked on a piece of the uncured fabric specified in 3.3.2, then cemented on the hood and cured with the hood after assembly. The printed labels shall be well defined and clearly legible after being tested for crocking, both wet and dry when tested as specified in 4.4.5. The format and contents of the instruction label shall be as follows:

INSTRUCTIONS FOR USE

1. Wear for protection against injury from dangerous liquid fuels and oxidizers.

2. DO NOT USE the same hood for fuels and oxidizers without washing the hood between usage.

3. Always keep adjustment straps snug and properly adjusted.

4. Wear with breathing apparatus which is placed over the coveralls. Don mask before putting on hood.

NOTE: Instruction 4 shall be applicable to class 2 hoods only.

3.3.7 Snap fasteners. Snap fasteners shall consist of a size 1 or 2, 24 line size button with socket for the female half and a size 1 or 2 eyelet with stud for the male half conforming to style 2, finish 2 of MIL-F-10884.

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3.3.8 Slides. The slides for adjustment straps shall be 3/4 + 1/64 inch (similar to North and Judd No. 02300) conforming to commercial standards and made from commercial brass, 14 or 16 gage, American Standard, with a black chemical finish conforming to requirements of MIL-F-495 (see 4.4.1.1).

3.3.9 Performance. The completed hood shall meet the specified requirements after vulcanizing.

3.3.9.1 Fuel resistance. The completed seams, patches and surrounding area shall show no evidence of cracking, stiffening, flaking, swelling, separation of coating from the fabric, separation of strapping from the seam, separation of patch from hood, or surface tack rating greater than II, when tested as specified in 4.4.5.

3.3.9.2 Acid resistance. The completed seams, patches, and surrounding area shall show no penetration, as indicated by change of color of the detector paper, no evidence of cracking, stiffening, flaking, separation of coating from fabric, separation of strapping from the seam, separation of patch from hood, or surface tack rating greater than II, when tested as specified in 4.4.5 (see 6.7).

3.3.9.3 Seam and seam strapping adhesion. Adhesion of the seam and seam strapping for all seams shall conform to requirements of table I when tested as specified in 4.4.5.

TABLE I. Requirements for seam and seam strapping adhesion

Test	Minimum requirement
Seam gum strapping adhesion	3.5 lb/inch
Seam gum strapping adhesion after immersion	3.0 lb/inch
Seam gum strapping after heat aging	3.0 lb/inch
Cemented seam adhesion	3.5 lb/inch
Cemented seam adhesion after immersion	3.0 lb/inch
Cemented seam adhesion after heat aging	3.0 lb/inch
Resistance to cold crack of finished cemented and strapped seam	1/
Seam hydrostatic resistance of finished cemented and strapped seam	70 psi

1/ The seam shall show no signs of cracking, flaking, separation, stiffening, delamination, or tackiness after test has been performed.

3.4 Design. The design of the hoods shall be as shown on Drawings 8-2-16 and 8-2-24 listed in 2.1.2, as applicable. The hoods shall be constructed and assembled with the heavily coated side of the cloth facing to the outside.

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3.5 Patterns. Standard patterns which provide the proper seam allowance for all seams will be furnished by the Government to the contractor for cutting working patterns. The Government patterns shall not be altered in any way and are to be used as a guide for cutting contractors working patterns. The working patterns shall be identical to the Government patterns.

3.5.1 List of pattern parts. The hood shall be cut from materials as specified in accordance with the pattern parts indicated in table II.

TABLE II. Pattern parts

Material	Nomenclature	Cut parts
	Class 1 hood	
Cloth, coated, nylon resin modified, butyl coated, acid and fuel resistant	Crown	1
	Front	1
	Adjustment strap	1
	Attachment strap	1
	Loop	1
	Facing	
Vinyl chloride sheet	Outside facepiece	1
	Inside facepiece	1
Cork composition	Outside facepiece reinforcement	1
	Inside facepiece reinforcement	1
Gum strapping compound	Front reinforcement	1
	Class 2 hood	
Cloth, coated, nylon resin modified, butyl coated, acid and fuel resistant	Front	1
	Crown and back	1
	Adjustment strap	2
	Attachment strap	2
	Loop	1
	Hood reinforcement patch	2
	Adjustment strap reinforcement	2
	Snap fastener reinforcement	2
	Facing	1
Vinyl chloride sheet	Outside facepiece	1
	Inside facepiece	1

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TABLE II. Pattern parts (cont'd)

Material	Nomenclature	Cut parts
Cork composition	Outside facepiece reinforcement	1
	Inside facepiece reinforcement	1
Gum strapping compound	Front reinforcement	1

3.6 Construction. The construction shall conform to table III and as specified herein.

3.6.1 Strapping of seams. All seams on the outer surface of the hood shall be sealed with the strapping specified in 3.3.3. After seam has been cemented, the strapping shall be centered and cemented over the raw edge of the seam on the outer surface of the hood.

3.6.2 Setting of snap fasteners. A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than two splits shall occur in the button or eyelet barrel.

3.6.3 Repairs in coating on finished hood. Areas containing coating defects may be repaired by application of a round patch 1-1/2 inches $\pm$ 1/4 inch in diameter using the strapping and front reinforcement gum strapping specified in 3.3.3 and the cement specified in 3.3.4, applied and cured as specified for the sealed seams in operations 8 and 18 of table III. The finished patch shall meet the acid resistance requirements specified in 3.3.9.2, when tested as specified in 4.4.5. The patch shall be cemented to the hood and the patch shall extend a minimum of 1/2 inch beyond all edges of the damaged area. The patch shall adhere smoothly and evenly over the entire area, and shall adhere well to the coated cloth so that, when subjected to flexing action between the hands, shall show no lifting of the edge, and no cracking, flaking, or removal of the cement or patch. The repaired area shall show no stiffening or tackiness which would affect the serviceability. Repairing of fabric coating defects by methods other than patching are not authorized.

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3.6.3.1 Location and number of repairs. Patches not exceeding four in number may be made to each hood, provided not more than two patches are made on any one of the two major pattern parts of the hoods. For the class 1 hood the parts applicable are the crown and front and for the class 2 hood the front and the crown - back parts.

3.7 Manufacturing operations requirements. The hoods shall be made in accordance with operation requirements specified in table III. The contractor is not required to follow the exact sequence of operations listed, except for operation 9, class 1 hood and operation 19, class 2 hood which shall be performed after the hood has been fully vulcanized.

3.8 Workmanship. The finished hood shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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TABLE III.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
	Class 1 Hood						
1.	Cutting.						
	a. Cut the material for hood in strict accordance with patterns which shows notches or marks for proper assembling of all parts.						
	b. Cut each part from one piece of material.						
	c. Cut the butyl rubber coated cloth so that the heavier coated side of the fabric shall form the outer surface of the hood.						
	d. Cut one hood loop reinforcement patch, 2 inches ± 1/8 inch in diameter.						
	e. Cut one adjustment strap reinforcement piece, 1-1/4 by 2-1/2 inches ± 1/8 inch.						
	f. Cut one snap fastener reinforcement piece, 1-1/4 inches ± 1/8 inch square.						
	g. Cut front reinforcement from gum strapping compound in accordance with pattern furnished (see 3.3.3).						
2.	Replacement of damaged parts.						
	a. Replace at time of cutting, any part containing a hole, cut, abraded area, tear, rip, area of no coating. brittle area which when bent results in a crack, embedded foreign material which when removed, leaves exposed basic fabric or a through hole.						

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NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of damaged parts. (cont'd) b. Replace any part damaged during the manufacturing process, such as a cut, tear, hole, mend, or drill hole.						
3.	Make loop and straps. a. Fold raw edges of cut pieces for loop and straps to center, lengthwise, overlapping one ply on the other, not less than 1/16 inch, and cement all plies together. b. The finished straps and loop shall measure as follows: (one each) (1) Adjustment strap: 26 inches ± 1/8 inch by 11/16 ± 1/16 inch. (2) Attachment strap: 4-1/2 inches ± 1/8 inch by 11/16 ± 1/16 inch. (3) Loop: 6 inches ± 1/8 inch by 3/8 ± 1/16 inch.						
4.	Make hood. a. Overlap crown on front 1/2 ± 1/8 inch, cement, and completely strap on the outside with gum strapping centered on raw edge of seam. b. Insert hood loop through cut opening in reinforcement patch (see 1.d.) with the ends spread open and cement in position. The loop shall protrude to the outside not less than 2 inches. c. Turn in raw edges at bottom, sides, and around crown, 3/8 ± 1/8 inch and cement to hood.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make hood. (cont'd) d. Cement the snap fastener reinforcement piece (see 1.f.) on front at right side at bottom of hood on outside surface as indicated by marks on pattern. e. Cement front reinforcement to face opening as indicated by marks on pattern with the edges of the reinforcement and face opening coinciding (see 1.g.). f. Cement facing to inside surface of hood along face opening except for top edges (forehead). Keep all edges evenly aligned. g. Cut back corners at top of face opening of hood and facing as indicated by marks on pattern. With inside surface of hood facing up, fold top edge of face opening back so that the fold back measures 1/4 ± 1/16 inch at center with a gradual taper to each end. Cement the folded edge down.						
5.	Attach adjustment and attachment straps. a. Position free end of the adjustment strap, 2 inches ± 1/8 inch, under a reinforcement patch and cement to inside of hood at left side as indicated by marks on pattern. b. Thread free end of strap over the center bar of slide, with the end, in turn, threaded through the top opening of a second slide and drawn under the center bar of the first slide, under the strap, with the free end inserted between the strap 1 ± 1/8 inch and cement on itself.						

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NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Attach adjustment and attachment straps. (cont'd)						
	c. Thread the attachment end of strap through bottom opening of the second slide. With ends even, cement the top and bottom plies of the strap together.						
6.	Attach snap fasteners (typical both straps).						
	a. Attach female snap fastener on attachment strap with button component on outside surface and with center of fastener located 5/8 ± 1/8 inch from end of strap.						
	b. Attach male snap fastener on right side of hood through hood and reinforcement piece with stud component of fastener on outside surface of hood (see 4.d.).						
7.	Attach label.						
	Stamp or cement label on inside of hood at center front, not less than 1 inch from bottom edge.						
8.	Dust and vulcanize hood.						
	The hood shall be dusted with an inert dusting powder and shall then be vulcanized.						
9.	Attach hood to frame.						
	a. Drill 14 holes around facepiece opening of hood, facepieces, and on the cork reinforcement pieces, as indicated by marks on pattern. The holes may be elongated not more than 1/4 inch, except on hood, to facilitate joining of layers.						

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TABLE III. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Attach hood to frame. (cont'd) (1) Position outside facepiece on outside of frame over face opening. Edges of facepiece shall be located at equal distances from inside edge of face opening of frame around entire periphery, ± 1/16 inch. Drill 14 holes through frame in same location as on facepiece. Hole size shall be 5/32 inch. b. Position layers of materials (from outside to inside) as follows: (1) Outside facepiece, (2) hood, (3) cork reinforcement, (4) frame, (5) cork reinforcement, and (6) inside facepiece. Attach layers using the screw and nut attachments (see 3.3.1.5), with the nut portion on the inside of the frame. The hood shall overlap the outside of the facepiece at the top edge of opening only, and the top edge of the facing on the inside shall be inserted between the frame and the cork reinforcement.						
10.	Check hood. Check all outside seams for security of strapping and seam coverage.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
	Class 2 Hood						
11.	Cutting.						
	a. Cut the material for hood in strict accordance with patterns furnished which show notches for proper assembling of all parts.						
	b. Cut each part from one piece of material.						
	c. Cut the butyl rubber coated cloth so that the heavier coated side of the fabric shall form the outer surface of the hood.						
	d. As required, cut reinforcement patches (size as indicated in operation 1).						
12.	Replacement of damaged parts.						
	a. Replace at time of cutting any part containing a hole, cut, abraded area, tear, rip, area of no coating, brittle area which when bent, results in a crack, embedded foreign material which when removed, leaves exposed basic fabric, or a through hole.						
	b. Replace any part damaged during the manufacturing process, such as a cut, tear, hole, mend, or drill hole.						
13.	Make loop and straps.						
	a. Fold raw edges of cut pieces for the loop and the straps to center, lengthwise, overlapping one ply on the other, not less than 1/16 inch and cement all plies together.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Make loop and straps. (cont'd) b. The finished straps shall measure as follows: (1) Adjustment strap: 2 each, 15 inches ± 1/8 inch by 11/16 ± 1/16 inch. (2) Attachment strap: 2 each, 4-1/2 inches ± 1/8 inch by 11/16 ± 1/16 inch.						
14.	Make hood. a. Insert each end of hood loop through cut opening in reinforcement patches with ends laying flat in slit opening and cement securely to patches. The free length from cut opening to cut opening shall measure 3 inches ± 1/4 inch long. Cement reinforcement patches to crown as indicated by marks on pattern. b. Overlap crown on front 5/8 + 1/8 inch, cement, and completely strap on the outside with gum strapping centered on raw edge of seam. c. Cement facing to inside surface of hood along face opening, except for top edges (forehead). Keep all edges evenly aligned. d. Cut back corners at top of face opening of hood and facing as indicated by marks on pattern. With inside surface of hood facing up fold top edge of face opening back so that the fold back measures 1/4 ± 1/16 inch at center with a gradual taper to each end. Cement the folded edge down.						

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NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Make hood. (cont'd) e. Cement front reinforcement to face opening as indicated by marks on pattern with the edges of the reinforcement and face opening coinciding. f. Turn in raw edges of sides of the front and back bibs 3/8 ± 1/8 inch and cement. g. Turn bottom edges of hood bibs at front and back to the inside 3/4 ± 1/8 inch and cement. h. Cement the snap fastener reinforcement pieces on back of each side at bottom of hood as indicated by marks on patterns.						
15.	Attach adjustment and attachment straps. a. Position free ends of the adjustment straps 2 inches ± 1/8 inch under a reinforcement patch and cement to inside of hood at both sides of bottom. Turn up on front. b. Thread free end of strap over the center bar of slide, with the end, in turn, threaded through the top opening of a second slide and drawn under the center bar of the first slide under the strap with the free end inserted between the straps 2 inches ± 1/8 inch and cement on itself. c. Thread the attachment end of strap through bottom opening of the second slide. With ends even, cement the top and bottom plies of the strap together.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Attach snap fasteners. a. Attach female snap fasteners on attachment straps with button component on outside surface and with center of each fastener located 5/8 ± 1/8 inch from end of each strap. b. Attach male snap fasteners on bottom back at each side of hood through hood and reinforcement piece with stud component of fastener on outside surface of hood.						
17.	Attach label. Stamp or cement label on inside of hood at center front, not less than 1 inch from bottom edge.						
18.	Dust and vulcanize hood. The hood shall be dusted with an inert dusting powder and shall then be vulcanized.						
19.	Attach hood to frame. a. Drill 14 holes around facepiece opening of hood, facepieces, and on the cork reinforcement pieces, as indicated by marks on pattern. The holes may be elongated not more than 1/4 inch, except on the hood, to facilitate joining of layers.						

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NO.	TABLE III. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Attach hood to frame. (cont'd) (1) Position outside facepiece on outside of frame over face opening. Edges of facepiece shall be located at equal distances from inside edge of face opening of frame around entire periphery, ± 1/16 inch. Drill 14 holes through frame in same location as on facepiece. Hole size shall be 5/32 inch. b. Position layers of materials (from outside to inside) as follows: (1) Outside facepiece, (2) hood, (3) cork reinforcement, (4) frame, (5) cork reinforcement, and (6) inside facepiece. Attach layers using the screw and nut attachments (see 3.3.1.5) with the nut portion on the inside of the frame. The hood shall overlap the outside of the facepiece at the top edge of opening only, and the top edge of the facing on the inside shall be inserted between the frame and the cork reinforcement.						
20.	Check hood. Check all outside seams for security of strapping and seam coverage.						

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.4.1.1 Certification. The contractor shall furnish, to the contracting officer, a certificate of compliance stating that the frame material, gum strapping, front reinforcement and slide conform to the requirements specified in section 3.

4.4.2 In-process inspection. Inspection of the hood and its components shall be made during the manufacturing operations to ascertain that construction details which cannot be examined on the finished product are in accordance with specified requirements. The in-process inspection shall include inspection of the working patterns to assure that they conform to the standard patterns, inspection of the cut parts covered by patterns to ascertain that the parts are cut properly with respect to size and orientation such that the heavy coated side of the coated fabric is on the exterior side of the hood, and that the diameter of the holes for the snap fasteners is smaller than the outside diameter of the button and eyelet barrels. Whenever nonconformance is noted, corrections shall be made to the components affected and lot in-process. Parts which cannot be corrected shall be removed from production.

NOTE: Determination of the heavy coated side of the fabric can be accomplished by noting which side the twill lines of the base fabric can be distinguished through The twill lines are distinguishable on the light coated side of the fabric only.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of one hood. The sample unit shall be one hood. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot.

TABLE IV. End item visual defects

Examine	Defect	Classification Critical	Major	Minor
Finish (applicable to all components)	Not as specified			X
Cleanness	Not clean (excessive adhesive adjacent to edge of seam not applicable)			X
	Any spot, stain, or foreign matter, clearly noticeable			X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Material (general)	Any component not fabricated of the specified material		X	
Coated fabric	Hole, cut, abraded area, tear, or rip through material or exposing base fabric	X		
	Area of no coating	X		
	Blister, tunnel, or delamination of coating	X		
	Brittle area which when bent results in a crack	X		
	NOTE: If any of the above four conditions occur on a reinforcement patch, hem, or adjustment strap, it shall be classified as a minor defect.			
	Area of excessive coating, i.e., solid runs, ridges, or lumps			X
	Patches or repairs in basic material.			
	Patch diameter less than 1-1/4 inches		X	
	Patch diameter greater than 1-3/4 inches			X
	More than two patches per pattern part			X
	More than four patches per hood		X	
	Edge of patch lifting or cracking, flaking or removal of the cement or patch when hand flexed as specified in 3.4.3	X		
	Patch not smoothly and evenly adhered		X	
	Patch area excessively stiff or tacky affecting serviceability	X		

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Coated fabric (cont'd)	Patch extends less than 1/2 inch beyond all edges of damaged area		X	
	Any unauthorized repair, e.g., cement applied to defect in fabric coating in lieu of patching		X	
	Pit or abraded area but not through material or not to a degree where base fabric is exposed			X
	Tacky, i.e, improper vulcanization		X	
	Embedded foreign material which when removed will expose base fabric or leave a through hole	X		
	Foreign material which cannot be readily removed or if removable, does not expose base fabric		X	
	NOTE: A pick glass may be used to determine the degree of the deficiency.			
	Any component omitted		X	
	Any component damaged, not as specified, or otherwise defective		X	
	Heavy coated side not on exterior of hood	X		
	Any component misplaced, not serving intended purpose	X		
	Any component insecurely attached or affixed	X		
	Any part twisted or distorted		X	
	Any exposed drill hole	X		
	Any part not cut in accordance with directional lines on patterns		X	
	Not dusted with powder			X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Assembly of vulcanized and cemented parts	Open seam completely through hood	X		
	Area of delamination other than on the strapping, but does not constitute an opening completely through hood			X
	Area of delamination at edge of strapping 1/16 inch or more, but does not extend to the raw edge of the seam			X
	Area of delamination of strapping extending to the raw edge of the seam	X		
	Inside edge of seam open more than 1/8 inch but not extending to gum strapping			X
	Seam strapping off center on outer edge of seam lap by more than 1/4 inch			X
	Seam strapping not covering raw edge of seam	X		
	Any blister in the strapping		X	
	Any hole, cut, tear, or rip in the seam strapping, exposing base fabric	X		
	NOTE: Extent of delamination (or tunnel) may be determined by insertion of any blunt end instrument suitable for the purpose. Penetration of the instrument shall cease immediately if resistance is met.			

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Hardware	Any part of snap fastener, slide, nut, or screw missing		X	
	Any slide malformed or damaged to a degree that it does not perform intended function		X	
	More than two screw-nut assemblies loosely joined			X
	Any screw-nut assembly excessively tight resulting in damage to facepiece		X	
	Position of nut and screw reversed, i.e., nut appears on the outside of hood			X
	Any part of snap fastener reversed		X	
	Any sharp burrs			X
	Any fastener not functioning properly, i.e., fails to snap closed, provide a secure closure, or open freely		X	
	Any fastener clinched excessively tight, cutting adjacent material		X	
	Any fastener clinched loosely, permitting either component to rotate freely		X	
	Incorrect style fastener		X	
	Three or more splits in eyelet or button barrel			X
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
Facepieces	Not optically clear		X	
	Any permanent stain seriously affecting optical clarity		X	
	Any hole, cut, crack, crazing, or cloudiness		X	
	Edges not smooth			X

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Face opening	Either cork reinforcement piece missing on inside or outside of hood		X	
	Hood overlaps facepiece at top and bottom in lieu of top only			X
Adjustment straps, attachment strap, and loop	Not located as specified		X	
	Free end of adjustment strap not threaded through slides as specified		X	
	Any reinforcement patch missing			X
	Hood loop missing		X	
Frame	Not type specified		X	
	Frame does not include means for adjusting head sizes		X	
	Damaged		X	
Marking (label)	Missing, incomplete, incorrect, not legible, not specified type or size, not in proper location, or not accomplished in the specified manner		X	

4.4.4 End item critical defect examination. Every end item in lots found acceptable under the visual examination in 4.4.3 shall be examined for the critical defects listed in table IV. Any end item found to contain one or more critical defects shall be rejected.

4.4.5 Testing of the end item. The method of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. The results of the physical testing of the coated materials, obtained under testing conditions defined in FED-STD-406, FED-STD-601, or FED-STD-191 will be acceptable except in dispute cases; in dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191. Except for the adhesion of gum strapping and cemented seams and the hydrostatic resistance of cemented seams, all the physical and chemical values specified in section 3 apply to results of each individual

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determination. The adhesion of gum strapping and cemented seams and the hydrostatic resistance of cemented seams value specified in section 3 apply to lot average of all determinations. The sample unit shall be four finished class 1 hoods or three finished class 2 hoods. 1/ In addition, for each sample unit, a specially prepared panel 6 yards long and 8 inches wide, finished with the sealed seam centered and running parallel to the long dimension shall be submitted for class 2 hoods. The specially prepared test pieces shall be made up of the same material and fabricated in the same manner as the end item. The frame, facepieces, and cork reinforcements shall not be part of the sample unit for testing. The sample size shall be as follows:

Lot size	Sample size (number of sample units)
280 or less	1
281 to 500	2

The lot shall be unacceptable if any individual determination or the lot average of all determinations, as applicable, fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

1/ Additionally, if any hoods contained in the lot have been repaired by patching (see 3.6.3), they are required to be segregated (see footnote 3/ to table V), and the sample units shall be selected from the patched hoods. If only one patched hood is contained in the lot, it shall be used for performing the applicable patch test.

TABLE V. End item tests

Component	Characteristic	Requirement paragraph	Test procedure	No. of determinations	Results reported as
Labels	Crocking, dry and wet	3.3.6	5651 1/	1	Pass or fail
Gum strapping	Adhesion	3.3.9.3	4.5.1	3	Numerically to nearest 0.1 lb/in.
	Adhesion after immersion	3.3.9.3	4.5.1.1	3	Numerically to nearest 0.1 lb/in.
	Adhesion after heat aging	3.3.9.3	4.5.1.2	3	Numerically to nearest 0.1 lb/in.

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TABLE V. End item tests (cont'd)

Component	Characteristic	Requirement paragraph	Test procedure	No. of determinations	Results reported as
Seams 2/	Adhesion	3.3.9.3	4.5.2	3	Numerically to nearest 0.1 lb/in.
	Adhesion after immersion	3.3.9.3	4.5.2.1	3	Numerically to nearest 0.1 lb/in.
	Adhesion after heat aging	3.3.9.3	4.5.2.2	3	Numerically to nearest 0.1 lb/in.
	Resistance to cold crack	3.3.9.3	4.5.3	1	Pass or fail
	Acid resistance 3/	3.3.9.2	4.5.4.3	2	Pass or fail
	Fuel resistance	3.3.9.1	4.5.4.4	2	Pass or fail
	Surface tack after fuel exposure	3.3.9.1	4.5.4.5	2	Pass or fail
	Surface tack after acid exposure	3.3.9.2	4.5.4.5	2	Pass or fail
	Hydrostatic resistance	3.3.9.3	5512 4/	5	Numerically to nearest 1.0 psi
Patches 5/	Acid resistance 3/	3.3.9.2	4.5.4.3	1	Pass or fail

1/ The label shall be cut from the hood to give a test specimen 4 by 8 inches with the legend centrally located and with the printing perpendicular to the lengthwise direction and so positioned within the specimen as to be encompassed in the test area.

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2/ At least one specimen shall be taken from the end item for initial adhesion tests (see 4.5.1 and 4.5.2) and acid and fuel resistance tests (see 4.5.4.3 and 4.5.4.4). Test specimens for other test characteristics will be taken from the end items as available. For the class 2 hood, other specimens may be taken from the specially prepared seam strip (see 4.4.5). If test piece and results indicate a significant difference between the test piece and the end item, then on subsequent lots the sample unit size shall consist of five end items only.

3/ The acid resistance test shall be performed by the government (see 6.7).

4/ Hydrostatic pressure shall be applied to the outside of the seam with seam centered directly over the opening.

5/ Applicable only when the lot contains hoods which have been patched (see 3.6.3).

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. Bulged or distorted container.
Content	Number of unit packs per shipping container is more or less than required.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Seam gum strapping adhesion test. The adhesion of the gum strapping to the seam shall be determined as specified in Method 5962 of FED-STD-191, except as follows: The test sample shall age at least 48 hours after application of the strapping and vulcanization of the assembly before testing. Only straight seamed and strapped sections shall be tested. The test specimen shall have the gum strapping peeled back from the seam for a distance of 2-1/2 to 3 inches. The end of the separated gum strapping shall be clamped in the upper jaw, and the lapped seam, with strapping removed, shall be clamped in the lower jaw. The results shall be calculated to the nearest 0.1 pound per inch of width.

4.5.1.1 Seam strapping adhesion after immersion. The strapping shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at 80° ± 9°F, removed from water, and while still dripping wet, tested for seam strapping adhesion as outlined in 4.5.1.

4.5.1.2 Seam strapping adhesion after heat aging. Samples shall be exposed for 7 days in a circulating air oven at 158° ± 2°F, removed from the oven, conditioned for 24 hours at standard conditions, and then tested for seam strapping adhesion as outlined in 4.5.1.

4.5.2 Cemented seam adhesion test. The samples used in the performance of the test specified in 4.5.1 (with gum strapping removed) shall be used as the test samples for the test. The adhesion of the cemented seam shall be determined as specified in Method 5960 of FED-STD-191. The results shall be calculated to the nearest 0.1 pound per inch of width

4.5.2.1 Seam adhesion after immersion. The seam shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at 80° ± 9°F, removed from the water, and while still dripping wet, tested for seam adhesion as outlined in 4.5.2.

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4.5.2.2 Seam adhesion after heat aging. Samples shall be exposed for 7 days in a circulating air oven at 158° ± 2°F, removed from the oven, conditioned for 24 hours at standard conditions, and then tested for seam adhesion as outlined in 4.5.2.

4.5.3 Resistance of finished seams to cold crack test. The test shall be performed as specified in Method 5874 of FED-STD-191, with the following exceptions: test temperature shall be -40° ± 2°F. Test specimens shall be conditioned for not less than 4 hours at test temperature prior to testing. Test specimens shall be prepared by folding so that two seam surfaces are superimposed. The roller weight shall be passed along the fold and across the seam. The test specimen shall be visually examined for failure after test. No hydrostatic test shall be performed after cold test.

4.5.4 Chemical resistance of finished seam test.

Precautionary Note: Because of the toxicity of propellant nitric acid and JP-4, the chemical resistance testing operations should be reviewed by appropriate authorities to ensure adequate ventilation, personal protective equipment, and work practices are used.

4.5.4.1 Apparatus and chemicals.

a. Test jig No. 2 conforming to Drawing 8-1-3
b. Wax - paraffin wax or a mixture of paraffin and beeswax containing not more than one-third beeswax
c. Petroleum jelly
d. Aluminum block - dimensions approximately 4 by 2 7/16 inches by 1/2 inch
e. Chemical agents detector paper - M9 conforming to MIL-P-51493
f. Propellant-nitric acid types IIIA or IIIB conforming to MIL-P-7254 (see 6.7)
g. Glass supporting and cover plates measuring 8 by 8 inches
h. Distilled water
i. Manifold paper conforming to type V of UU-P-328
j. Roller - approximately 5-1/2 inches in diameter, 2 inches wide, and 10 pounds in weight
k. Calcium chloride-anhydrous conforming to type II, grade A of O-C-105

4.5.4.2 Preparation of test specimen. The test specimen shall be a section, 6 inches in diameter, including a length of finished strapped seam equal to the specimen diameter, or when testing patches, the patch shall be approximately in the center of the specimen, when possible. The test specimen shall be placed in the test jig in the following manner:

a. Place the test specimen between the spacer inserts and seal with melted wax. The wax shall form a continuous bead covering the outer circumference of the insert assembly.

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b. Place the insert assembly in the test jig sleeve and fill the cavity between the insert assembly and the sleeve wall with melted wax. Mate the jig sleeve and jig collar and screw hand tight.

c. Seal the inside of the sleeve collar assembly with melted wax. The wax seal shall be continuous and shall extend around and out onto the test specimen approximately 1/8 inch, and up the inside of test jig to a point approximately 1/8 inch above the top of the spacer insert. Turn the jig assembly over and seal the bottom cavity with melted wax in similar fashion.

d. During the chemical exposure, the top of the test apparatus shall be covered with a glass plate.

4.5.4.3 Acid resistance of the finished seam or patch and surrounding area (see 6.7).

a. Prepare test specimen as specified in 4.5.4.2.

b. Place test assembly in desiccator over calcium chloride conforming to type II, grade A of O-C-105 and condition at 70° ± 2°F, for a minimum of 16 hours.

c. After conditioning, remove from desiccator, and immediately perform acid exposure test as follows:

(1) Coat bottom and top edges of the test jig with 1/32 inch of petroleum jelly.

(2) Place aluminum block in the center of a glass supporting plate with a piece of vesicant detector paper, coated side up, on top of the block. Place the test assembly over the block with the heavily coated side of the test specimen up.

(3) Cover the test specimen with 50 milliliters of propellant nitric acid and cover top of the test jig with glass cover plate.

d. After 2 hours acid exposure at 70° ± 2°F and 65 ± 2.0 percent relative humidity, remove the test specimen from the test jig, rinse in distilled water, dry at stated conditions for approximately 4 hours and examine for requirements in 3.3.9.2.

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4.5.4.4 Fuel resistance of the finished seam. The test specimen, prepared as specified in 4.5.4.2 and conditioned as specified in 4.5.4.3, shall be covered with 50 milliliters of JP-4 (MIL-T-5624) for 15 minutes at conditions of 70° ± 2°F, and 65 ± 2.0 percent relative humidity. After fuel exposure, the test specimen shall be removed from the test jig, rinsed in distilled water, dried for approximately 4 hours at the stated conditions, and examined for requirements in 3.3.9.1.

4.5.4.5 Surface tack rating of the finished seam. The surface tack test shall be conducted, under conditions of 70° ± 2°F, and 65 ± 2.0 percent relative humidity, as follows:

a. The test specimen, prepared as specified in 4.5.4.2 and conditioned as specified in 4.5.4.3, shall be covered with 50 milliliters of the specified acid or fuel.

b. After 15 minutes exposure, under conditions of 70° ± 2°F and 65 ± 2.0 percent relative humidity, the specimen shall be removed from the test jig, rinsed in distilled water, and dried for approximately 4 hours at stated conditions.

c. Surface tack rating shall be determined as follows:

(1) The specimen, with wax line removed by cutting or scraping, shall be placed exposed side up on a smooth hard surface, and covered with a 2 by 4 inch piece of manifold paper.

(2) The roller shall be passed along the full length of the manifold paper covered seam twice in each direction (four passes) with pressure applied by the weight of the roller only.

(3) After rolling, the manifold paper shall be peeled from the test specimen and the surface tack rating determined by comparison with figure 1 and rating standards.

Rating standards: 1/

Rating I - A rating of I shall be equivalent to no trace 2/ of coating material retained on the test paper or no trace of paper fiber adhering to the test specimen surface.

Rating II - A rating of II shall be equivalent to a trace 2/ of coating material retained on the test paper or a trace of paper fiber adhering to the test specimen surface.

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Rating III - A rating of III shall be equivalent to any degree of coating material or paper fiber transfer in excess of rating II.

1/ Figure 1 pictorially represents the Rating Standards.

2/ A trace is defined as coating material or paper fiber transfer covering 25 percent of test specimen area.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each hood shall have the adjustable head band turned down toward the facepiece. The loose coated fabric sections of the hood shall be carefully tucked into the well of the facepieces. Each hood shall be inserted into a flat or square style clear polyethylene film bag conforming to A-A-50083. Each bagged hood shall be unit packed in a fiberboard box (facepiece squarely facing the bottom of the box) conforming to style RSC, type CF (variety SW), class domestic, grade 200 of PPP-B-636. Inside dimensions of each fiberboard box shall be approximately 14-1/2 inches in length, 9-1/4 inches in width, and 7-1/2 inches in depth. Approximate dimensions are furnished as a guide only. The top and bottom flaps of each box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. Hoods shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Eight hoods of one class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, two in length, two in width, and two in depth, within a shipping container. Inside dimensions of the shipping container shall be approximately 30 inches in length, 19-1/2 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Eight hoods of one class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Level A unit packs shall be packed flat, two in length, two in width, and two in depth within a shipping container. Inside dimensions of the shipping container shall be approximately 30 inches in length, 19-1/2 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Hoods, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), hoods, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 1 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The hood is intended for wear by military personnel engaged in handling acid and fuel propellants as a component of the Coverall, Rocket Fuel Handler's, Impermeable, Limited Protection.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Class required (see 1.2).
c. When a first article sample is required (see 3.2, 4.3 and 6.4).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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6.3 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Frame. A source of supply for the frame (part no. 622) is Fibre Metal Products Co., Brinton Lake Rd. and Rte. 1, Concordville, PA 19331.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

* 6.7 Performance of acid resistance tests. Because of the danger involved in handling and shipping of propellant nitric acid, the acid resistance test specified in 4.5.4.3 will be performed by the Government. The samples to be tested for acid resistance should be forwarded to the following address:

Commander
Defense Personnel Supply Center
Clothing and Textile Laboratory Branch
ATTN: DPSC-FQL
2800 South 20th Street
Philadelphia, PA 19101

* 6.8 Subject term (key word) listing.

Hood
Protective clothing
Rocket fuel handlers

* 6.9 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.10 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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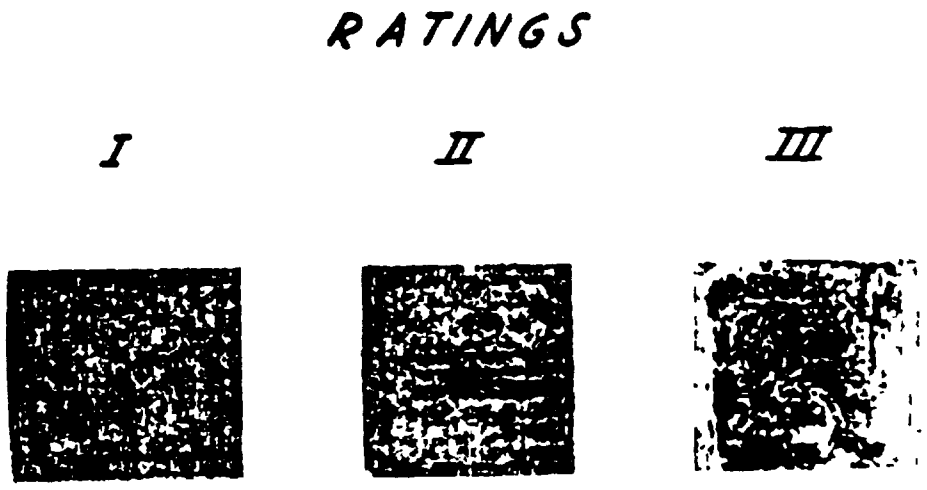


FIGURE 1. Surface tack standards (nitric acid test).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-H-43079F	Hood, Rocket Fuel Handler's: Impermeable, Limited Protection

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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